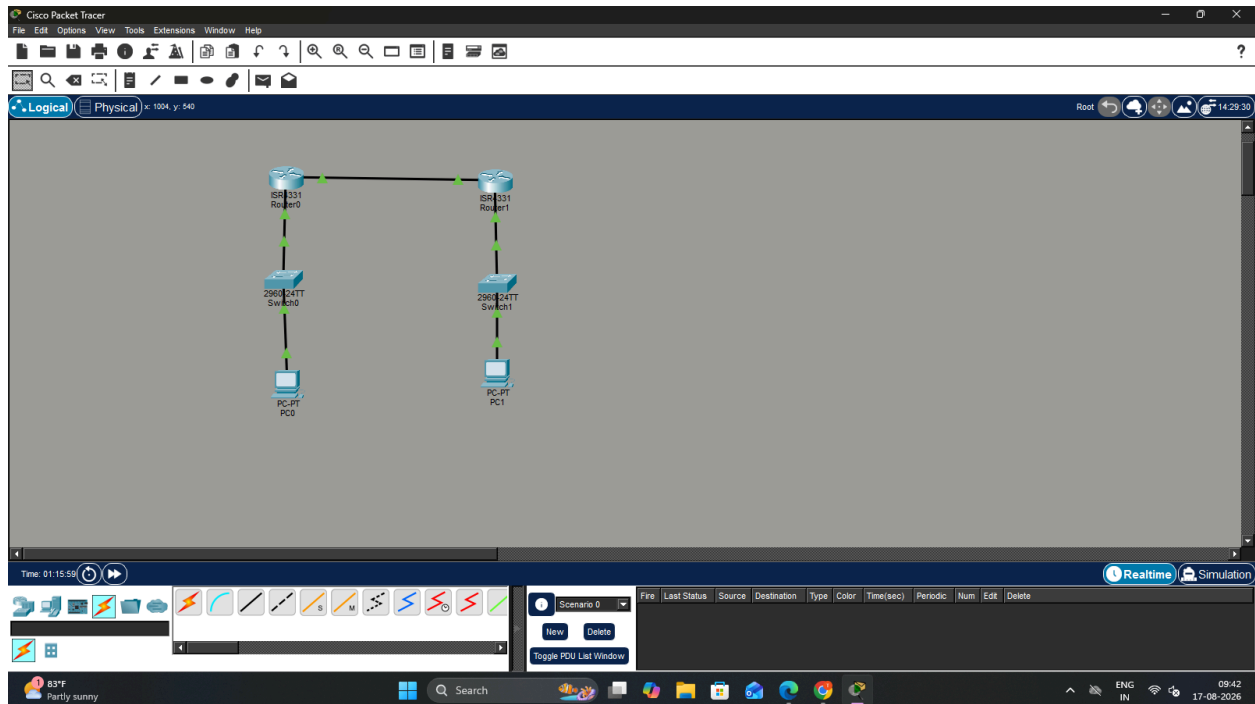


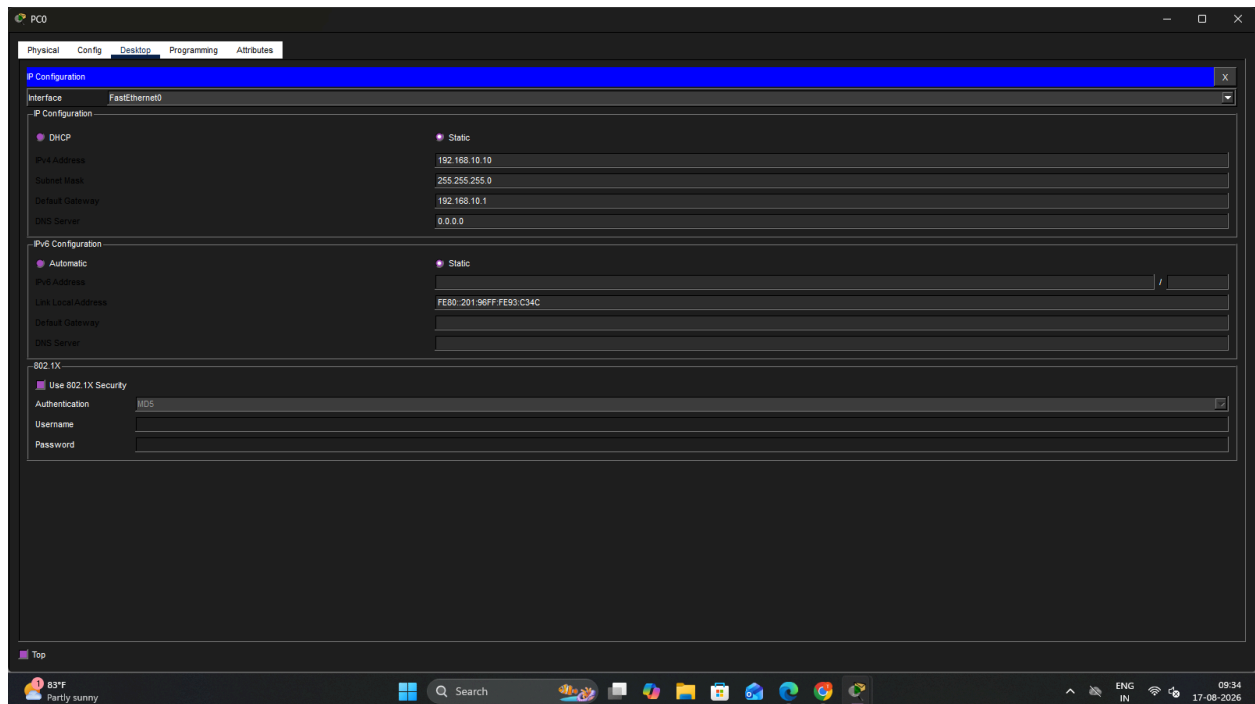
PRACTICAL NO:6

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Aim :- Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.



Pc 0

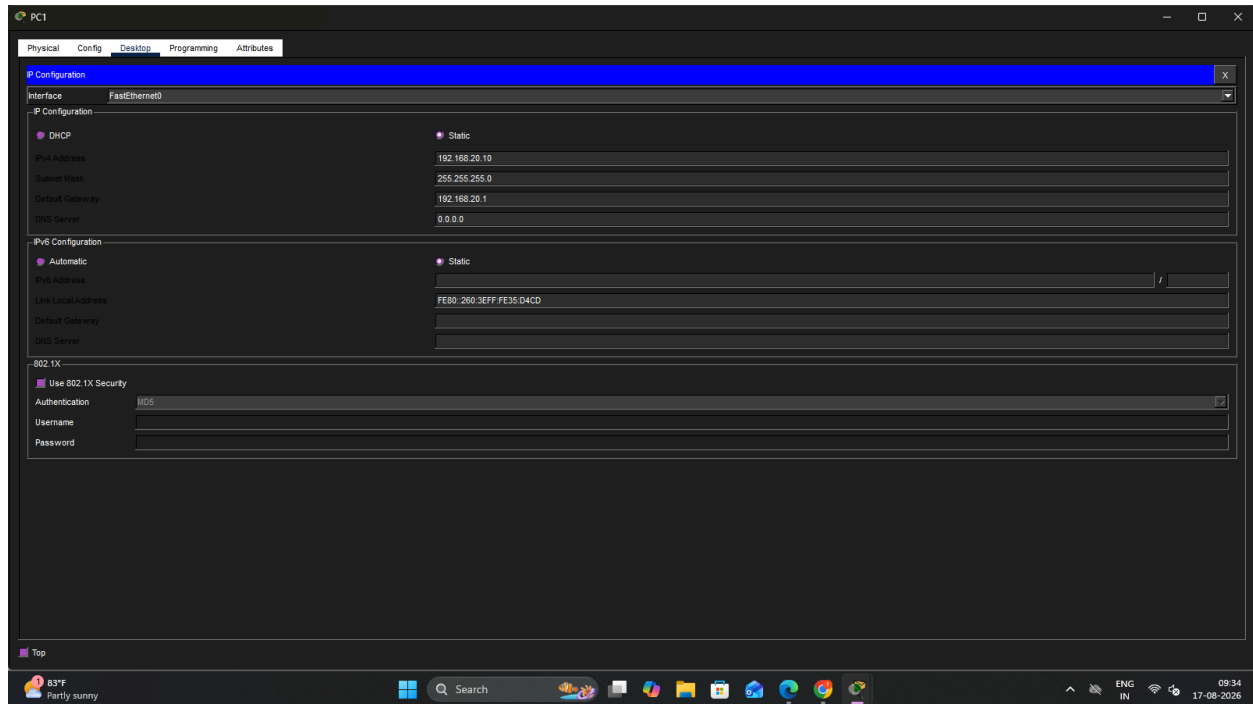


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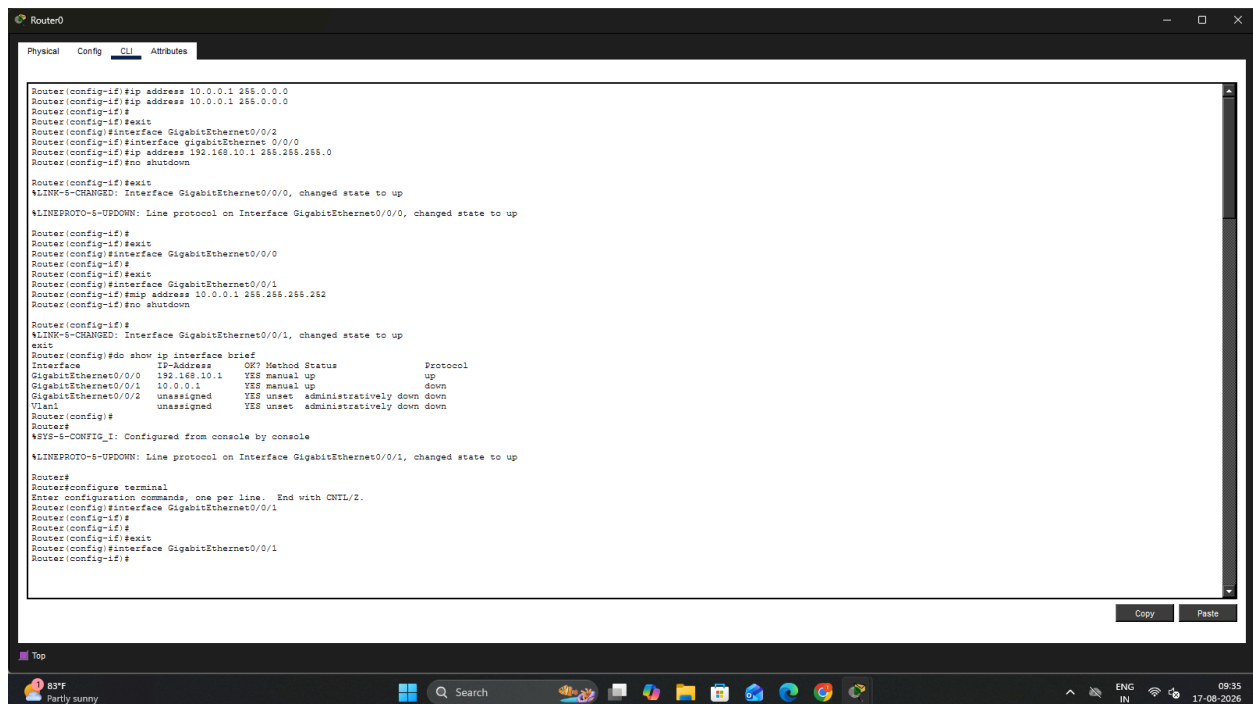
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Pc 1



Router 0



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Router0

Physical Config CLI Attributes

```

Router>enable
Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 10.0.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encr aes 256
Router(config-isakmp)#hash sha256
^
* Invalid input detected at '^' marker.

Router(config-isakmp)#
Router#
*SYS-5-CONFIG_I: Configured from console by console

Router#hash sha
^
* Invalid input detected at '^' marker.

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#hash sha
Router(config-isakmp)#encr aes 256
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 14
^
* Invalid input detected at '^' marker.

Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router#
*SYS-5-CONFIG_I: Configured from console by console

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#group 5
Router(config-isakmp)#crypto isakmp key cisco123 address 10.0.0.2
^
* Invalid input detected at '^' marker.

```

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Top

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Router0

Physical CLI Attributes

```

Router(config)#crypto isakmp key cisco123 address 10.0.0.2
Router(config)#crypto ipsec transform-set T5 esp-aes 256 esp-sha-hmac
Router(config)#crypto map CHAP 10 ipsec-isakmp
! NOTE: This new crypto map will remain disabled until a peer
      and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.2
Router(config-crypto-map)#set transform-set T5
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#ip access-list extended 100
Router(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config-ext-nacl)#exit
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#crypto map CHAP
*Jan 3 07:16:16.785: %CRYPTO-6-ISAHSH_ON_OFF: ISAHSH is ON
Router(config-if)#exit
Router(config)#end
Router#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...

[OK]

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, I - IS
       i - IS-IS, L1 - IS-IS Level-1, L2 - IS-IS Level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/10 is directly connected, GigabitEthernet0/0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0/0

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#end
Router#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...

[OK]

```

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Search

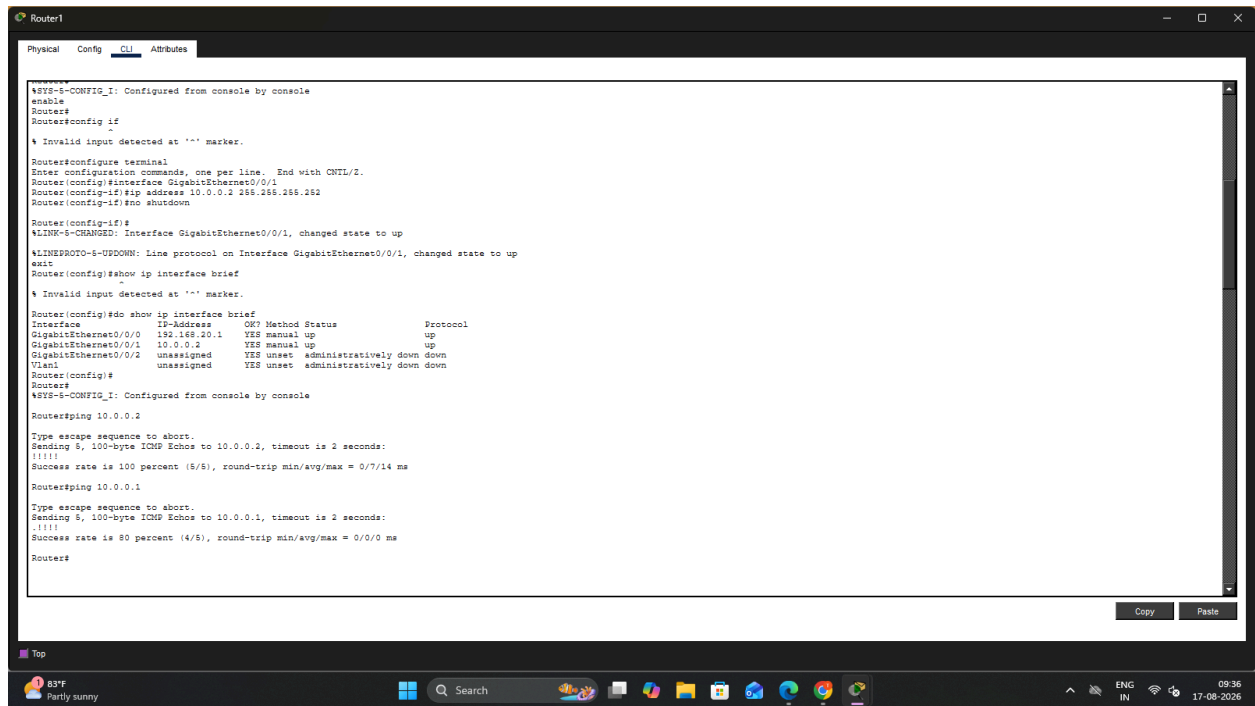
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```
Router1
Physical Config CLI Attributes

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
Router(config)#show ip interface brief
% Invalid input detected at '^' marker.

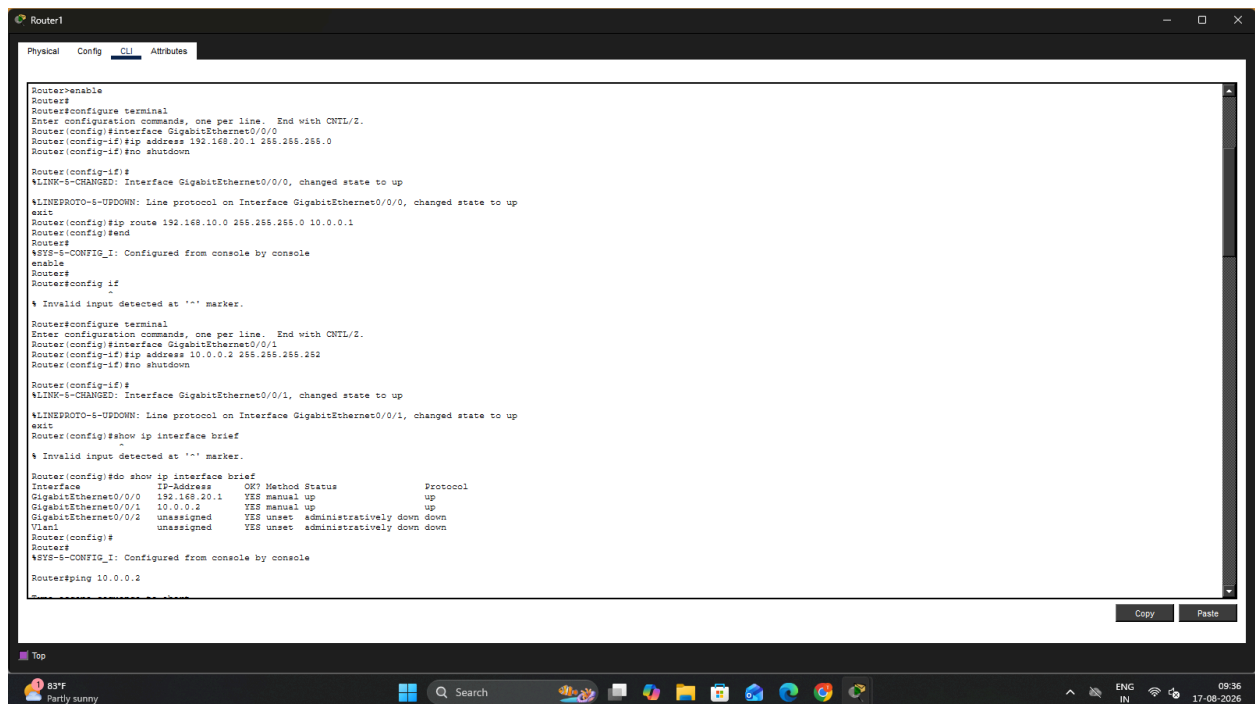
Router(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.20.1 YES manual up up
GigabitEthernet0/0/1 10.0.0.2 YES manual up up
GigabitEthernet0/0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/7/14 ms

Router#ping 10.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Router 1



```
Router1
Physical Config CLI Attributes

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.10.0 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
Router(config)#show ip interface brief
% Invalid input detected at '^' marker.

Router(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.10.0 YES manual up up
GigabitEthernet0/0/1 10.0.0.2 YES manual up up
GigabitEthernet0/0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

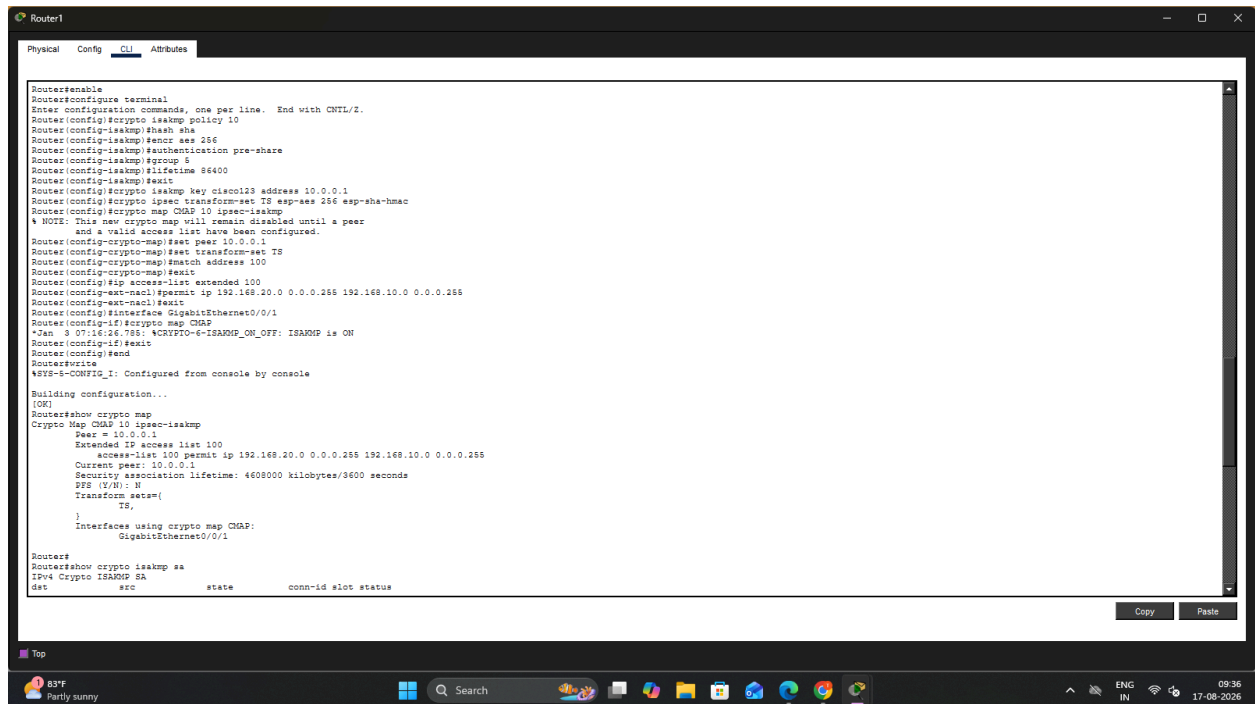
Router#ping 10.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

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PRACTICAL NO:6

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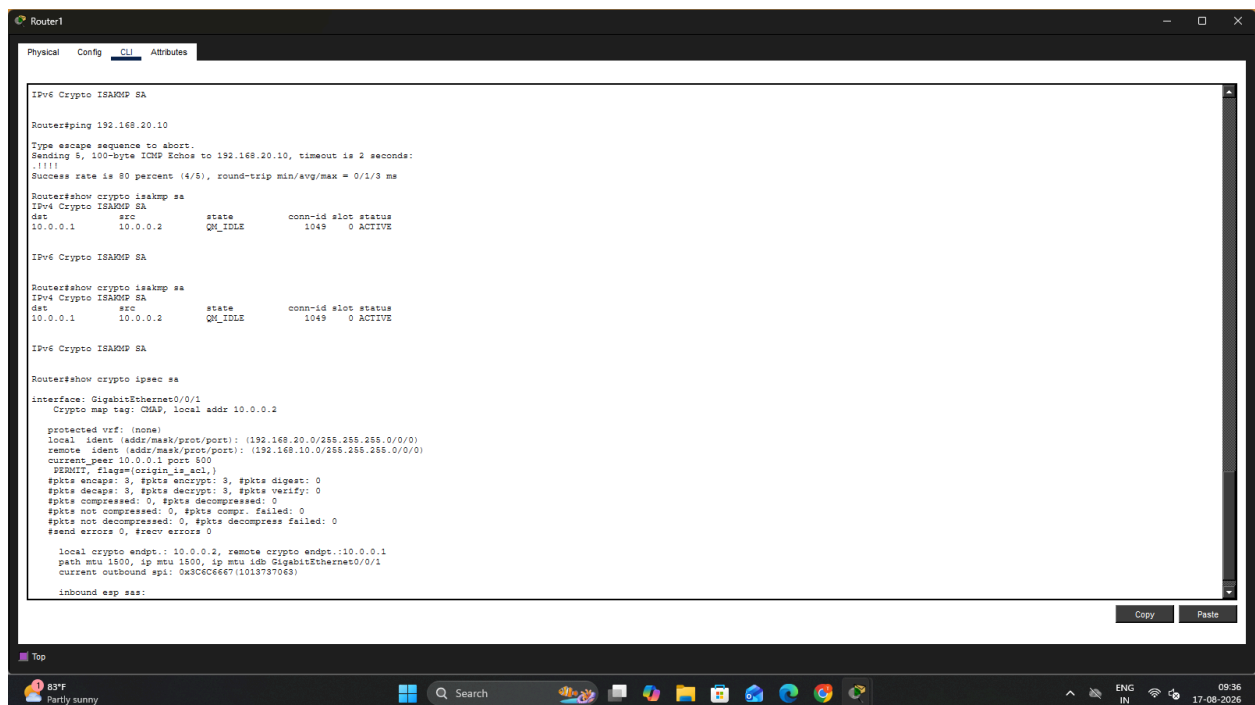


```
Router1
Physical Config CLI Attributes

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#hash sha
Router(config-isakmp)#encr aes 256
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#crypto ipsec transform-set TS esp-aes 256 esp-sha-hmac
Router(config)#crypto map CHAP 10 ipsec-isakmp
! NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 192.168.20.0
Router(config-crypto-map)#set transform-set TS
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#ip access-list extended 100
Router(config-ext-nacl)#permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config-ext-nacl)#exit
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#crypto map CHAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#end
Router#write
$SYS-S-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#show crypto map
Crypto Map CHAP 10 ipsec-isakmp
  Peer = 192.168.20.0
  Extended IP access list 100
    access-list 100 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
  Current peer: 192.168.20.0
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (V/M): N
  Transform set=
    TS
  Interfaces using crypto map CHAP:
    GigabitEthernet0/0/1

Router#
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.1     10.0.0.1     QM_IDLE        1049         0 ACTIVE
```



```
Router1
Physical Config CLI Attributes

IPv6 Crypto ISAKMP SA

Router#ping 192.168.20.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 192.168.20.10, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/1/3 ms

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.1     10.0.0.2     QM_IDLE        1049         0 ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.1     10.0.0.2     QM_IDLE        1049         0 ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto ipsec sa
interface: GigabitEthernet0/0/1
Crypto map tag: CHAP, local addr 10.0.0.2

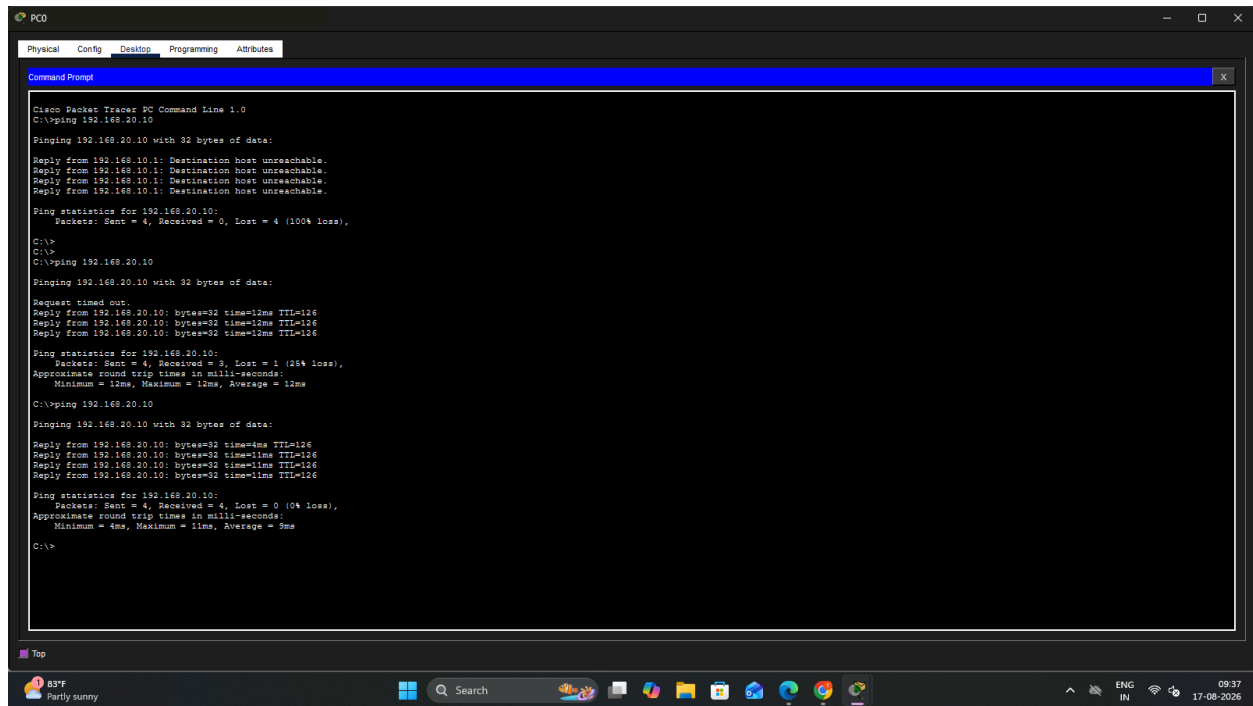
protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
current_peer 10.0.0.1 port 100
PERMIT, flags={origin_is_acl,}
#pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
#pkts decaps: 3, #pkts decrypt: 3, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr: failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 10.0.0.2, remote crypto endpt.: 10.0.0.1
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0/1
current outbound spi: 0ad3c066e71013737063

inbound esp sas:
```

PRACTICAL NO:6

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```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
C:\>
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=12ms TTL=126
Reply from 192.168.20.10: bytes=32 time=12ms TTL=126
Reply from 192.168.20.10: bytes=32 time=12ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 12ms, Average = 12ms
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 11ms, Average = 9ms
C:\>
```